

Mechanics Of Materials
Basic Mech. Engg.

Mechanics Of Materials
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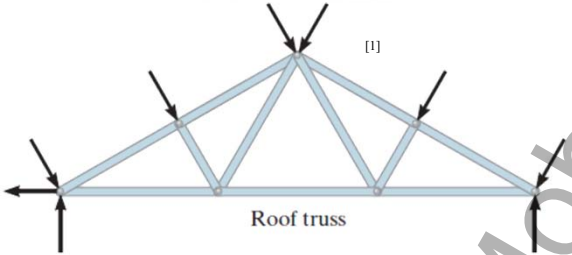
Lloads
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[1]



Roof truss

If the entire truss is in equilibrium, then each of its joints is also in equilibrium



[1] - Russell C. Hibbeler. Engineering mechanics: Statics and dynamics 12th edition SI, Fig 6-1, pp. 263

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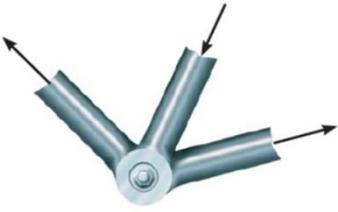
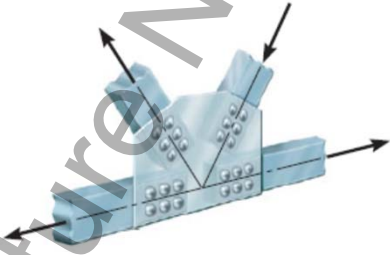
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Force equilibrium equations can be used to obtain the member forces acting on each joint

[1] - Russell C. Hibbeler. Engineering mechanics: Statics and dynamics 12th edition SI, Fig 6-3(a, b), pp. 264

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Method of Joints

PROCEDURE FOR ANALYSIS

- FREE BODY DIAGRAM**
 Draw the free-body diagram of a joint having **at least one known force** and **at most two unknown forces**. (If this joint is at one of the supports, then it may be necessary first to calculate the external reactions at the support.)
- EQUATIONS OF EQUILIBRIUM**
 - Orient the x and y axes such that the forces on the free-body diagram can be easily resolved into their x and y components and then apply the two force equilibrium equations and Solve for the two unknown member forces and verify their correct sense.
 - Using the calculated results, continue to analyze each of the other joints. Remember that a member in compression “pushes” on the joint and a member in tension “pulls” on the joint. Also, be sure to choose a joint having at most two unknowns and at least one known force.

[1] - Russell C. Hibbeler. Engineering mechanics: Statics and dynamics 12th edition SI, pp. 267
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Loads

INTERNAL LOAD IN TRUSSES – METHOD OF JOINTS

Always start at a joint having at least one known force and at most two unknown forces

[1] - Russell C. Hibbeler. Engineering mechanics: Statics and dynamics 12th edition SI, Example 6.1, pp. 268
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INTERNAL LOAD IN TRUSSES – METHOD OF JOINTS

The diagram shows a truss structure with a vertical member AB of height 2 m and a horizontal member AC of length 2 m. A horizontal load of 500 N is applied at joint B. The diagonal member BC is at a 45° angle to the horizontal. The right diagram shows the internal forces at the joints: at joint B, the vertical member is in tension (500 N) and the diagonal member is in compression (707.1 N); at joint A, the horizontal member is in tension (500 N) and the vertical member is in tension (500 N); at joint C, the horizontal member is in tension (500 N) and the diagonal member is in compression (707.1 N).

[1] - Russell C. Hibbeler. Engineering mechanics: Statics and dynamics 12th edition SI, Example 6.1, pp. 268

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INTERNAL LOAD IN TRUSSES – METHOD OF JOINTS

The diagram shows a truss structure with a horizontal member AD of length 6 m and a vertical member CD of height 4 m. A horizontal load of 600 N is applied at joint D. A vertical load of 400 N is applied at joint B. The diagonal member BC is at a 45° angle to the horizontal. The right diagram shows the internal forces at the joints: at joint B, the vertical member is in tension (400 N) and the diagonal member is in compression (500 N); at joint A, the horizontal member is in tension (600 N) and the vertical member is in tension (400 N); at joint C, the horizontal member is in tension (600 N) and the diagonal member is in compression (500 N).

[1] - Russell C. Hibbeler. Engineering mechanics: Statics and dynamics 12th edition SI, Example 6.3, pp. 270

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INTERNAL LOAD IN TRUSSES – METHOD OF JOINTS

The diagram shows a truss structure with a horizontal base of 6 m and a vertical height of 4 m. A downward load of 400 N is applied at the top-left joint. At the top-right joint, there is a horizontal reaction force C_x and a vertical reaction force C_y . At the bottom-left joint, there is a vertical reaction force A_y . At the bottom-right joint, there is a horizontal reaction force D_x and a vertical reaction force D_y . The internal forces at the joints are analyzed as follows:

- Joint A:** A free-body diagram shows a vertical force of 600 N acting upwards, a horizontal force F_{AD} acting to the right, and a diagonal force F_{AB} acting upwards and to the right. A slope triangle indicates a vertical component of 4 and a horizontal component of 3.
- Joint B:** A free-body diagram shows a horizontal force F_{CB} acting to the left, a vertical force of 200 N acting downwards, and a horizontal force of 600 N acting to the right.
- Joint C:** A free-body diagram shows a horizontal force F_{CB} acting to the left, a vertical force of 200 N acting downwards, and a horizontal force of 600 N acting to the right.
- Joint D:** A free-body diagram shows a horizontal force of 450 N acting to the left, a vertical force of 600 N acting upwards, and a diagonal force F_{DB} acting upwards and to the left. A slope triangle indicates a vertical component of 4 and a horizontal component of 3.

[1] - Russell C. Hibbeler. Engineering mechanics: Statics and dynamics 12th edition SI, Example 6.3, pp. 270

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INTERNAL LOAD IN TRUSSES – METHOD OF JOINTS

The diagram shows the same truss structure as above, but with internal forces labeled as Compression or Tension. The forces are as follows:

- Member AB:** Compression, 750 N.
- Member BC:** Compression, 600 N.
- Member CD:** Compression, 200 N.
- Member DA:** Tension, 450 N.
- Member BD:** Tension, 250 N.

[1] - Russell C. Hibbeler. Engineering mechanics: Statics and dynamics 12th edition SI, Example 6.3, pp. 271

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INTERNAL LOAD IN TRUSSES – METHOD OF JOINTS

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Loads

ZERO-FORCE MEMBERS [1]

- Zero-Force members are those members that support **no loading**.
- They are used to **increase the stability** of the truss during construction and to provide **added support** if the loading is changed.

How to identify a zero-force member?

FBD at Joint A

$$\begin{aligned} \rightarrow \sum F_x &= 0; F_{AB} = 0 \\ \uparrow \sum F_y &= 0; F_{AF} = 0 \end{aligned}$$

FBD at Joint D

$$\begin{aligned} +\curvearrowright \sum F_y &= 0; F_{DC} \sin \theta = 0; \\ +\rightarrow \sum F_x &= 0; F_{DE} + 0 = 0; \\ F_{DC} &= 0 \text{ since } \sin \theta \neq 0 \\ F_{DE} &= 0 \end{aligned}$$

[1] - Russell C. Hibbeler. Engineering mechanics: Statics and dynamics 12th edition SI, pp. 272

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Loads

ZERO-FORCE MEMBERS [1]

- Zero-Force members are those members that support **no loading**.
- They are used to **increase the stability** of the truss during construction and to provide **added support** if the loading is changed.

How to identify a zero-force member?

If only two members form a truss joint and no external load or support reaction is applied to the joint, the two members must be zero-force members.

[1] - Russell C. Hibbeler. Engineering mechanics: Statics and dynamics 12th edition SI, pp. 272

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ZERO-FORCE MEMBERS [1]

How to identify a zero-force member?

FBD at Joint D

FBD at Joint C

$$+\curvearrowleft \sum F_x = 0; \quad F_{DA} = 0$$
$$+\curvearrowright \sum F_y = 0; \quad F_{DC} = F_{DE}$$

$$+\curvearrowleft \sum F_x = 0; \quad F_{CA} \sin \theta = 0;$$
$$+\curvearrowright \sum F_y = 0; \quad F_{CB} = F_{CD}$$
$$F_{CA} = 0 \text{ since } \sin \theta \neq 0;$$

[1] - Russell C. Hibbeler. Engineering mechanics: Statics and dynamics 12th edition SI, pp. 273

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ZERO-FORCE MEMBERS [1]

How to identify a zero-force member?

If three members form a truss joint for which two of the members are collinear, the third member is a zero-force member provided no external force or support reaction is applied to the joint.

[1] - Russell C. Hibbeler. Engineering mechanics: Statics and dynamics 12th edition SI, pp. 273

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ZERO-FORCE MEMBERS [EXAMPLE]

Collinear Rule

$F_{GC} = 0$

$F_{DC} = 0$

$F_{DF} = 0$

$F_{FC} \cos \theta = 0$

Since $\theta \neq 90^\circ$

$F_{FC} = 0$

[1] - Russell C. Hibbeler. Engineering mechanics: Statics and dynamics 12th edition SI, Example 6.4, pp. 274

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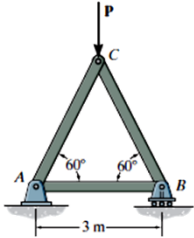
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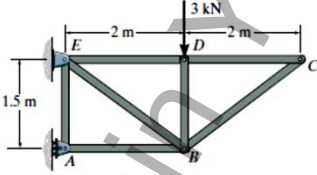
Method of Joints

FUNDAMENTAL PROBLEMS

F6-4. Determine the greatest load P that can be applied to the truss so that none of the members are subjected to a force exceeding either 2 kN in tension or 1.5 kN in compression.



F6-5. Identify the zero-force members in the truss.



[1] - Russell C. Hibbeler. Engineering mechanics: Statics and dynamics 12th edition SI, pp. 286

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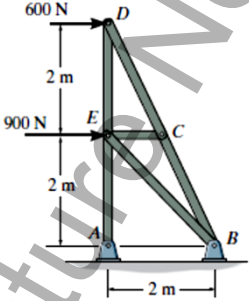
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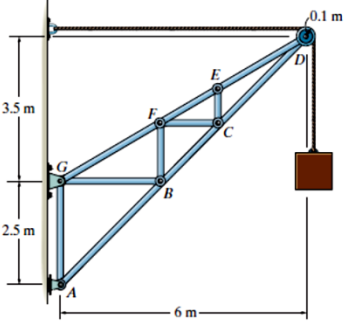
Method of Joints

PROBLEMS

•6-1. Determine the force in each member of the truss, and state if the members are in tension or compression.



•6-21. Determine the largest mass m of the suspended block so that the force in any member does not exceed 30 kN (T) or 25 kN (C).



[1] - Russell C. Hibbeler. Engineering mechanics: Statics and dynamics 12th edition SI, pp. 286

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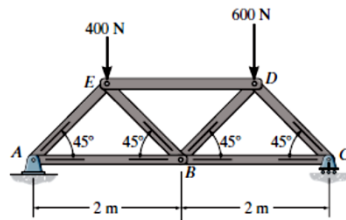
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Method of Joints

PROBLEMS

6–23. The truss is fabricated using uniform members having a mass of 5 kg/m. Remove the external forces from the truss, and determine the force in each member due to the weight of the truss. State whether the members are in tension or compression. Assume that the total force acting on a joint is the sum of half of the weight of every member connected to the joint.



[1] - Russell C. Hibbeler, Engineering mechanics: Statics and dynamics 12th edition SI, pp. 286